Vehicle Diagnostics Primer (synthetic sample)

Controller Area Network (CAN) is a serial bus that lets electronic control units in a vehicle exchange messages without a central host. Each message has an identifier that sets its priority, and a node that loses arbitration waits and retries.

A classical CAN frame carries up to eight data bytes. Typical bus speeds are 125 kbit/s for body electronics and 500 kbit/s for powertrain. Two twisted wires, CAN high and CAN low, carry a differential signal, and a 120 ohm terminating resistor sits at each end of the bus.

When a node detects an error it sends an error frame. A node with repeated errors moves to the error passive state and finally goes bus off, so that one faulty unit cannot block the whole network.

On-Board Diagnostics and Brakes (synthetic sample)

OBD-II is the standard diagnostic interface found on passenger cars. A scan tool connects to the 16 pin connector under the dashboard and reads diagnostic trouble codes. A code such as P0301 reports a cylinder 1 misfire. The malfunction indicator lamp turns on when the engine control unit stores an emissions related fault.

Hydraulic disc brakes press brake pads against a rotor. Brake fluid transfers pedal force to the calipers, and the fluid absorbs water over time, which lowers its boiling point. Anti-lock braking uses wheel speed sensors to release pressure when a wheel is about to lock.

Tire pressure monitoring systems warn the driver when a tire loses more than a set share of its pressure. Direct systems use a sensor inside each wheel, indirect systems compare wheel speeds.